SAFESIGHT / CIS-HW-01

EDGE AI | WORKPLACE SAFETY

One camera. A clearer view of risk.

A fixed-camera safety monitoring prototype that combines PPE and fire/smoke detections, restricted-zone activity, and explainable risk rules into one local dashboard.



SOFTWARE PROTOTYPE IN PROGRESS

Pitch brief | 05 November 2026 | GitHub: github.com/Cxen77/safesight

The problem and the proposition

Small teams often rely on people noticing safety issues in busy spaces. SafeSight explores whether one fixed camera and edge AI can help make those signals easier to see and review.

PPE signals

Detect people, visible heads, helmets, and vests. Flag missing-PPE evidence for review; appearance alone cannot establish compliance.

Zone activity

Use a configured polygon and tracked foot points to estimate entries, crowd count, and time spent in a restricted area.

Fire and smoke cues

Add a lightweight visual alert layer for fire/smoke-like detections. It is an early-warning cue, never a certified fire alarm.

PROPOSED VALUE

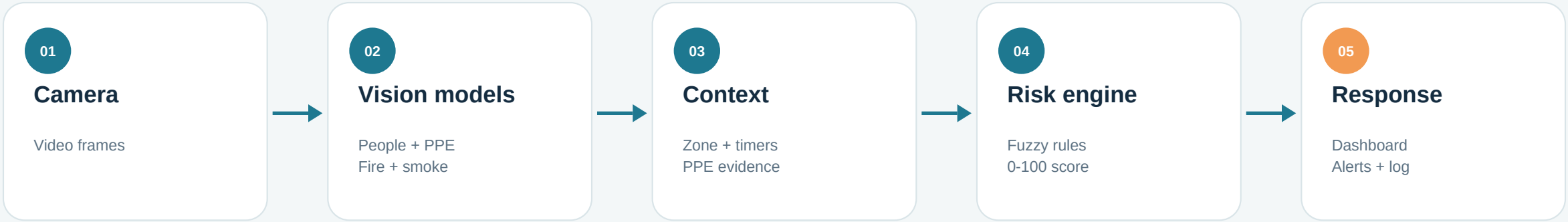
Turn separate visual cues into a single, reviewable risk signal with the reason, time, and event snapshot recorded locally.

This is a decision-support concept. It does not replace PPE policy, supervision, certified alarms, or emergency procedures.

How the system works

PROPOSED LIVE DATA PATH

The current code connects these stages. Full operation still depends on finished models, a camera-specific zone, and target-hardware checks.



EDGE-FIRST DEPLOYMENT GOAL

Process video on or near the camera location, so the safety workflow can continue locally and avoid a cloud dependency for the core loop. The laptop prototype is not yet proof of Jetson Nano performance.

Planned output: Low / Medium / High / Critical | local event snapshot and CSV log

From detections to a risk signal

A two-stage Mamdani fuzzy engine combines normalized inputs. Its rule base and monotonicity checks are documented in the repository.

PPE LOAD

min(1, (2 x missing helmets + missing vests) / (3 x people))
Empty scene = 0

ZONE PRESSURE

0.5 x min(intruders / 2, 1)
+ 0.5 x min(longest intrusion / 5 sec, 1)
No intruders = 0

PROXIMITY

Distance of tracked person feet to the configured hazard boundary, normalized to 0-1.

FIRE EVIDENCE

Smoothed fire/smoke model confidence from recent checks, normalized to 0-1.

FUZZY RISK BANDS

LOW	MEDIUM	HIGH	CRITICAL
0-25	26-50	51-75	76-100

Hard overrides: fire evidence >= 0.7 => Critical; observed unhelmeted intrusion >= 3 sec => at least High; >= 8 sec => at least Critical.

WHY USE FUZZY LOGIC?

Inputs are uncertain and graded. Membership functions and readable rules let the system combine weaker signals without pretending that every borderline case is perfectly certain.

The score is a prototype prioritization aid, not a probability or a safety certification.

Data and model development

PPE dataset: SH17

8,099 image/label pairs prepared. Seed-42 split: 5,669 train / 1,214 validation / 1,216 test. Classes: person, head, helmet, vest.

Fire dataset: D-Fire

21,501 valid prepared images across preserved train / validation / test splits. 26 invalid image-label pairs were excluded and recorded.

Training approach

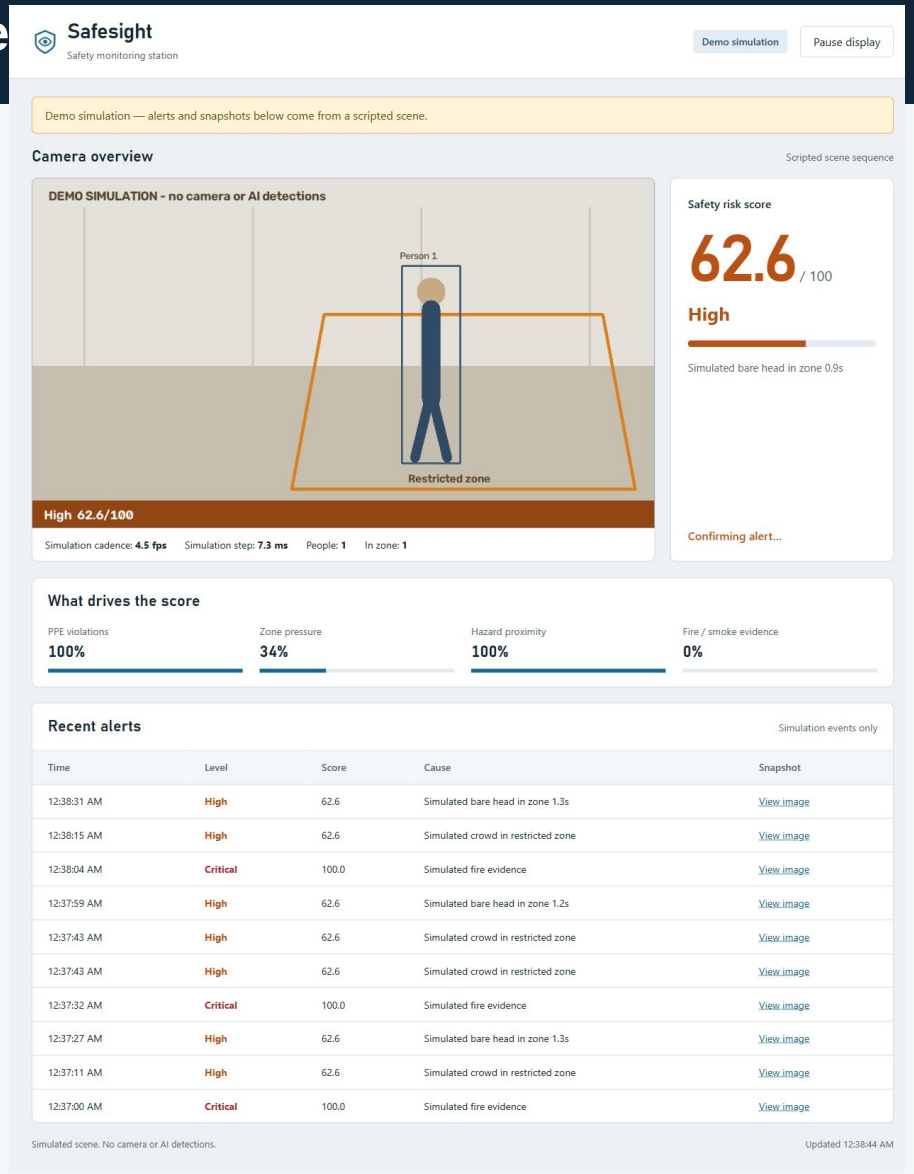
Start from a general COCO YOLOv8n checkpoint. Keep held-out tests frozen; report measured results only after final models are evaluated.

CURRENT EVIDENCE AND LIMITS

- PPE training stopped after 26 completed epochs; the log shows epoch 27 had begun. A resumable checkpoint is in the repo, but no final PPE model or frozen-test result is available.
- Fire training and its test evaluation have not started. The pipeline needs a recorded clip with a matching zone to measure end-to-end speed.
- No site photos are available, so the baseline cannot establish accuracy in the eventual demo surroundings. Dataset counts are not model accuracy.

SH17-derived weights carry CC BY-NC-SA 4.0 terms; see repository Credits before reuse or redistribution.

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PROTOTYPE FEATURES

- Risk score, band, and visible input contributions
- Restricted-zone overlay and people count
- Recent alerts with timestamps and event snapshots
- CSV event history and change-only alert output
- Explicit starting, unknown, stale, and error states

Important: this image is a scripted simulation. It uses no camera and no AI detections. The values shown are illustrative, not measured system results.

Where the project stands

AS OF 05 NOVEMBER 2026

IMPLEMENTED IN SOFTWARE

- Public dataset preparation, split checks, and frozen test snapshots
- Person/PPE inference, polygon-zone geometry, tracking timers, and crowd count
- 27 Stage 1 plus 9 Stage 2 fuzzy rules; monotonic logic verification
- Dashboard, event snapshots, CSV logging, alert policy, and GPIO output path
- Synthetic demo and automated evaluation/training scripts

No accuracy, real-time FPS, latency, or field-safety claim is made yet.

NOT COMPLETE / NEEDS ACCESS

- Finish PPE training, then evaluate once on the frozen public test set
- Train and evaluate fire/smoke model; publish the final measured weights
- Measure full pipeline on at least 60 seconds of recorded video
- Have a safety professional review 30 scenario labels; agreement is unknown
- Check real camera placement, zone drawing, and Jetson Nano GPIO/performance

THE NEXT 30 DAYS

From laptop prototype to a credible pilot.

1

Complete and measure

Resume PPE training; evaluate held-out data once; train and evaluate fire/smoke.

2

Validate the workflow

Capture a 60-second clip; measure end-to-end speed; get expert review of 30 scenarios.

3

Test on target setup

Configure the actual camera zone; verify alert outputs and Jetson Nano limits.

REQUEST: PILOT ACCESS + REVIEW

WHAT WE NEED FOR A PILOT

- Jetson Nano or intended edge device, camera, and GPIO access
- A permitted demo location and a defined restricted zone
- A short recorded clip from the intended environment
- A safety professional to review the scenario labels and threshold behavior

Repository: https://github.com/Cxen77/safesight